

1.FIND-S ALGORITHM

```
print("J THASLIMA NASREEN 192424152")
import pandas as pd
# Read CSV file
df = pd.read_csv("find s - Sheet1.csv")
# Rename the last column
df.rename(columns={df.columns[-1]: "Allergy"}, inplace=True)
print("Training Data:\n")
print(df)
# Ignore the Number column
concepts = df.iloc[:, 1:-1].values
target = df.iloc[:, -1].values
# Initialize hypothesis
hypothesis = ['0'] * len(concepts[0])
print("\nInitial Hypothesis:", hypothesis)
for i in range(len(target)):
    if target[i] == "Yes":
        if hypothesis == ['0'] * len(concepts[0]):
            hypothesis = list(concepts[i])

        else:
            for j in range(len(hypothesis)):
                if hypothesis[j] != concepts[i][j]:
                    hypothesis[j] = '?'

    print("\nAfter Positive Example", i + 1)
    print(hypothesis)

print("\nFinal Hypothesis")
print(hypothesis)
```

OUTPUT

Training Data:

	Number	Restaurant	Meal	Day	Cost	Allergy
0	Visit1	Sam's	Breakfast	Friday	Cheap	Yes
1	Visit2	Kim's	Lunch	Friday	Expensive	No
2	Visit3	Sam's	Lunch	Saturday	Cheap	Yes
3	Visit4	Bob's	Breakfast	Sunday	Cheap	No
4	Visit5	Sam's	Breakfast	Sunday	Expensive	No

Initial Hypothesis: ['0', '0', '0', '0']

After Positive Example 1

["Sam's", 'Breakfast', 'Friday', 'Cheap']

After Positive Example 3

["Sam's", '?', '?', 'Cheap']

Final Hypothesis

["Sam's", '?', '?', 'Cheap']